California State Polytechnic University Pomona

Aerospace Engineering

ARO4080

FEM Documentation

for

Test 4 Project

by

Jacob Harris

21 November 2025

Table of Contents

VOLUME I: FEM DOCUMENTATION REPORT

Table of Contents iii

CHP	DESCRIPTION	PAGE
1.	Introduction	1-1
	• Purpose	1-1
2.	FEM Documentation	2-1
	• Sketch of Model	2-1
	• Bulk Data File	2-2
	• Material Properties	2-3
	• Applied Loads	2-3
	• Constraints	2-4
3.	FEM Results	3-1
	• Nodal Displacements	3-1
	• SPC Forces	3-2
	• Grid Point Force Balance	3-2
	• Beam Axial Load, Shear, and Moment Diagrams in Frames	3-3
	• Beam Axial Load, Shear, and Moment Diagrams in Floor Beams	3-4
	• Beam Axial Load, Shear, and Moment Diagrams in Struts	3-5
4.	Conclusion	4-1
	• What I did/What it Means	4-1
	• What I Learned	4-1
	• Difficulty	4-1
	• Usefulness of Project	4-2
	• Effectiveness of FEMAP	4-2

APPX	DESCRIPTION	PAGE
A	FEM Preliminary Sketch	A-1
B	BDF of FEM	B-1

VOLUME II: F06 OUTPUT FILE

ARO4080_T-04-PJ-HarrisJacob.f06

VOLUME III: COMPLETE BDF LISTING

In Appendix B

1.0 Introduction

1.1 Purpose

The main purpose of this project was to build and analyze a finite element model of a single fuselage frame with dummy frames on both sides so we could see how an aircraft floor load actually flows through the structure. Instead of just looking at simple beams in isolation, this model ties together the frame, floor beams, skin, stringers, and struts and treats them as one system. For my case, the assigned fuselage has a 210 in diameter, 22 in frame spacing, floor struts at y = +- 66i n, and stringers every 15 degrees around the circumference.

There are a few main purposes for this project, the obvious one was to create a clean FEMAP model that follows all of the idealizations given in the test handout. That includes using CBAR elements for the frames and floor beams with the given areas and moments of inertias, CROD elements for the stringers between frames, CQUAD4 elements for the skin, and CBAR elements for the multi-segment struts that connect the floor back into the frame. The model also had to use the specified aluminum material properties and the "breathing wagon wheel" constraint system at the forward and aft dummy frames so the fuselage is supported but still allowed to expand radially. The purpose here is not just to get something that runs, but to show that we can translate a written structural idealization into a consistent FEM and document it clearly.

The second purpose is to simulate a realistic loading case and see how the structure reacts. In this project the load is a 1200 lb ultimate downward force spreading uniformly along the floor. The model was used to compute the axial forces, shears, bending moments, and deflections for the frame, floor, and strut members so we can draw internal load diagrams. The idea is to understand where the floor load goes and how much ends up in the floor beams themselves, how much transfers into the struts, and how the ring frame and stringers share the load with the skin.

Finally, main purpose of this assignment is just practice with the whole FEM workflow and documentation. The project forces us to start from a hand sketch, build the mesh, assign properties, loads, and constraints, and then back up the results with figures, force balances, and discussion instead of just screenshots. By the end, we should be able to explain not only what the maximum displacement or reaction was, but also why the load paths make sense for this kind of fuselage cross section and how confident we are in the model's behavior.

2.0 F.E.M. Documentation

2.1 Sketch of Model

Figures 2.1.1 and 2.1.2 show below depicts the FEMAP model of a single fuselage frame bay that I built for this project with one "dummy" frame on the fwd and aft sides. The frame rings are modeled with floor beams across the bottom, connected to the frame by diagonal struts, and all the frames are tied together by stringers. A 1200 lb or 5.92 lb/in uniform distributed load is applied along the floor beams of the main frame, while the "dummy" frames are contstrained with fixed "breathing wagon wheel" supports. Overall, the figures represent the full finite element setup that I used to study how the floor load flows into the fuselage skin.

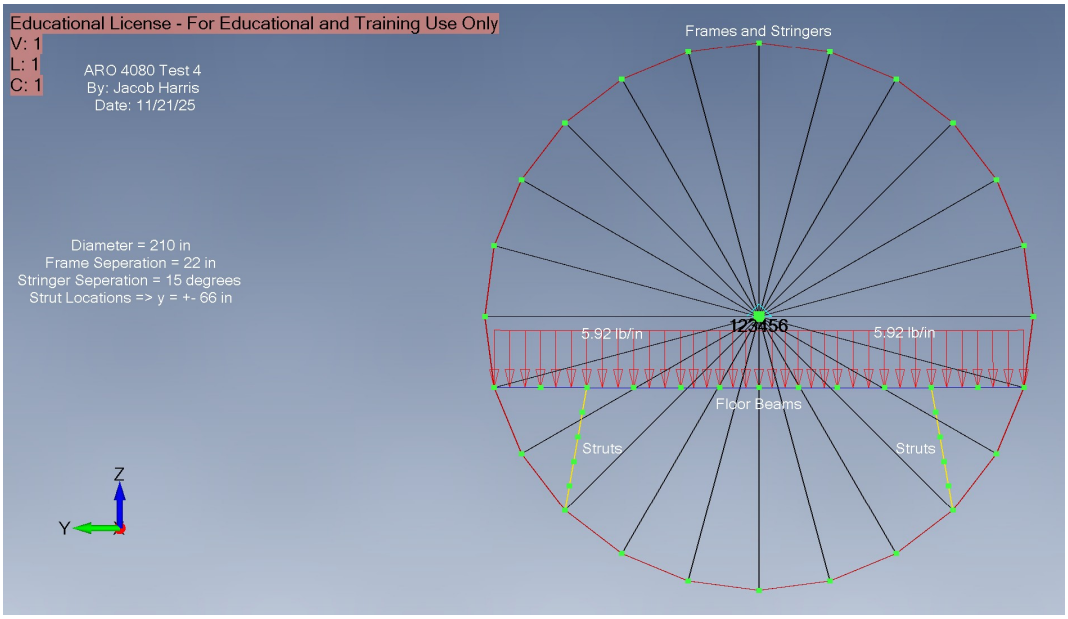


Figure 2.1.1: FEMAP FEM Cross Section View

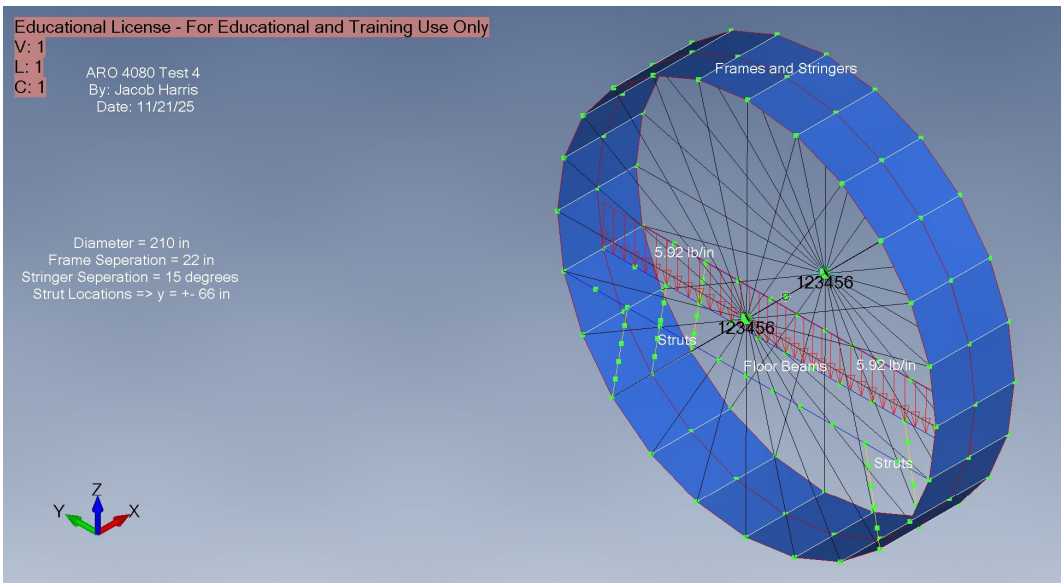


Figure 2.1.2: FEMAP FEM Isometric View

2.2 Bulk Data File

The FEM bulk data file (BDF) was generated from FEMAP after the finite element model was analyzed. It has all the necessary components to make this FEM run correctly. Due to the size, this BDF file can be found in Appendix B.

2.3 Material Properties

This FEM is constructed on only aluminum elements. This material is identified on a single MAT1 card, as follows:

```
MAT1,1,10.0E6,3.8e6            $ MAT1,id,E,G,v
```

The elastic & shear modulus for this material are shown. Poisson's ratio is computed by NASTRAN during runtime.

2.4 Applied Loads

There is a uniform Distributed Load of 12000 lb force or 5.92 lb/in applied on the floor beams of this FEM. It is important to note that this was only applied on the main frame and not the "dummy" frames, as shown on the PLOAD1 cards below.

```
   PLOAD1, id,Eid,Type,Scale,X1,P1,X2,P2
PLOAD1, 1, 51, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 52, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 53, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 54, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 55, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 56, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 57, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 58, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 59, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 60, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 61, FYE, FR,  0.,-5.92, 1.,-5.92
PLOAD1, 1, 62, FYE, FR,  0.,-5.92, 1.,-5.92
```

2.5 Constraints

This model used “Breathing Wagon Wheel” constraints on both dummy frames which allowed for a more realistic distribution of force from the floor beams to the skin. The spokes of the wheel are given tiny axial stiffness so the ring can “breathe” while still being stiff. Both wheels are fixed in the x,y,z axis’ for transverse translations (123) and rotational translations (456). The SPC1 cards for the FEM are shown below.

```
SPC1, id, C, G1
SPC1, 1, 123456, 74
SPC1, 1, 123456, 118
```

3.0 FEM Results

3.1 Nodal Displacements

The largest displacement according to the FEMAP results is going to be in the floor beam itself with a maximum deflection in the center of 0.1828 in. This depiction of the deflection is scaling the deflections so that we can see them clearly, we would likely not be able to tell otherwise.

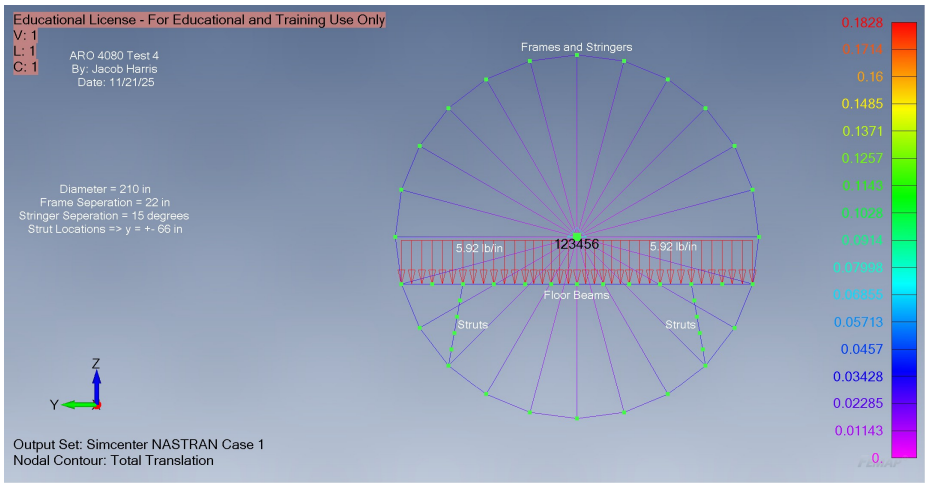


Figure 3.1.1: FEMAP FEM Undeformed Translation

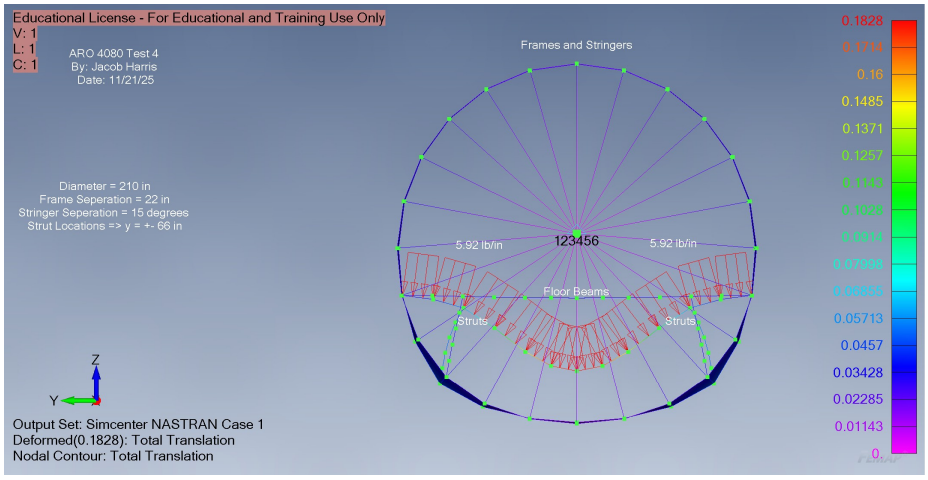


Figure 3.1.2: FEMAP FEM Deformed Translation

3.2 SPC Forces

The f06 FEMAP output describes the total vertical reaction is 1200 lb, as well as moments in x and z direction of 0.120 lb -in and 0.048 lb-in respectively. So the supports are carrying the entire load with negligible moments which is expected. The SPC forces are summarized below.

SPCFORCE RESULTANT

SUBCASE/ DAREA	ID	LOAD TYPE	T1	T2	T3	R1	R2	R3
0	1	FX	1.352873E-11	----	----	----	0.000000E+00	0.000000E+00
		FY	----	1.483975E-11	----	0.000000E+00	----	1.441888E-06
		FZ	----	----	1.200838E+03	0.000000E+00	-4.852831E-02	----
		MX	----	----	----	1.200838E-01	----	----
		MY	----	----	----	----	4.852831E-02	----
		MZ	----	----	----	----	----	-1.444608E-06
		TOTALS	1.352873E-11	1.483975E-11	1.200838E+03	1.200838E-01	-5.184120E-10	-2.720189E-09

3.3 Grid Point Force Balance

Here are the grid point force balances for the floor beams, for each node it lists how much of the applied load lands on the node (shown as APP-LOAD) and the internal forces/moments coming in from the connected beam elements (Bar Rows). The TOTALS line sums them together at each node.

G R I D P O I N T F O R C E B A L A N C E

	POINT-ID	ELEMENT-ID	SOURCE	T1	T2	T3	R1	R2	R3
0	26		APP-LOAD	0.0	0.0	-9.768000E+01	4.884000E+01	0.0	0.0
	26	56	BAR	-8.023463E-09	-2.016512E+02	-1.332000E+02	3.095503E+03	1.913593E-07	2.051187E-01
	26	57	BAR	8.023463E-09	2.016512E+02	2.308800E+02	-3.144343E+03	-1.913593E-07	-2.051187E-01
	26		*TOTALS*	3.330669E-16	-2.842171E-14	8.924417E-12	-2.910383E-11	-2.220446E-16	3.552714E-15
0	27		APP-LOAD	0.0	0.0	-8.880000E+01	5.684342E-14	0.0	0.0
	27	55	BAR	-8.023463E-09	-2.016512E+02	-4.440003E+01	5.093503E+03	1.913593E-07	2.051185E-01
	27	56	BAR	8.023463E-09	2.016512E+02	1.332000E+02	-5.093503E+03	-1.913593E-07	-2.051185E-01
	27		*TOTALS*	8.326673E-17	0.0	-7.389644E-13	2.910383E-11	0.0	5.884182E-15
0	28		APP-LOAD	0.0	0.0	-8.880000E+01	5.684342E-14	0.0	0.0
	28	54	BAR	-8.023463E-09	-2.016512E+02	4.439997E+01	5.759504E+03	1.913593E-07	2.051184E-01
	28	55	BAR	8.023463E-09	2.016512E+02	4.440003E+01	-5.759504E+03	-1.913593E-07	-2.051184E-01
	28		*TOTALS*	-3.690624E-16	0.0	-1.143263E-11	-4.001777E-11	0.0	6.522560E-15
0	29		APP-LOAD	0.0	0.0	-8.880000E+01	5.684342E-14	0.0	0.0
	29	53	BAR	-8.023462E-09	-2.016512E+02	1.332000E+02	5.093504E+03	1.913593E-07	2.051183E-01
	29	54	BAR	8.023463E-09	2.016512E+02	-4.439997E+01	-5.093504E+03	-1.913593E-07	-2.051183E-01
	29		*TOTALS*	6.188626E-16	0.0	-9.308110E-13	-9.094947E-12	2.220446E-16	-4.024558E-15
0	31		APP-LOAD	0.0	0.0	-9.768000E+01	-4.884000E+01	0.0	0.0
	31	53	BAR	8.023462E-09	2.016512E+02	-1.332000E+02	-3.095505E+03	-1.913593E-07	-2.051182E-01
	31	60	BAR	-8.023462E-09	-2.016512E+02	2.308800E+02	3.144345E+03	1.913593E-07	2.051182E-01
	31		*TOTALS*	-1.387779E-16	2.842171E-14	1.256240E-11	4.547474E-11	-2.220446E-16	-5.107026E-15

3.4 Axial Load, Shear, Moment Diagrams in Frames

The max Axial, Shear, and Moments are 382 lb, 142 lb, and -3126 in-lb respectively, located at the point where the struts meet the frames.

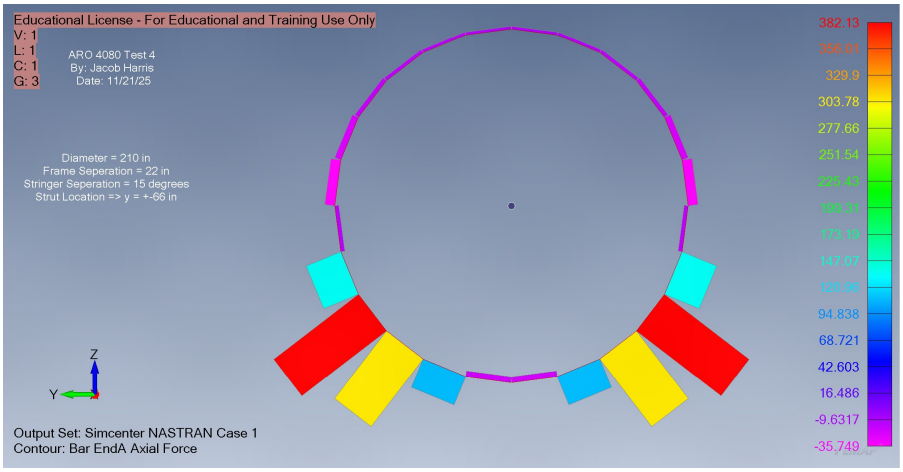


Figure 3.4.1: FEMAP FEM Axial Force in Frames

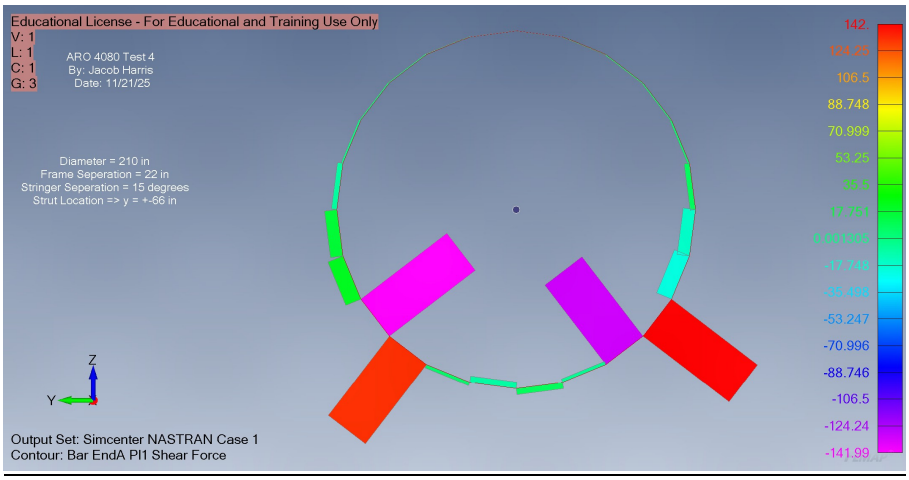


Figure 3.4.2: FEMAP FEM Shear in Frames

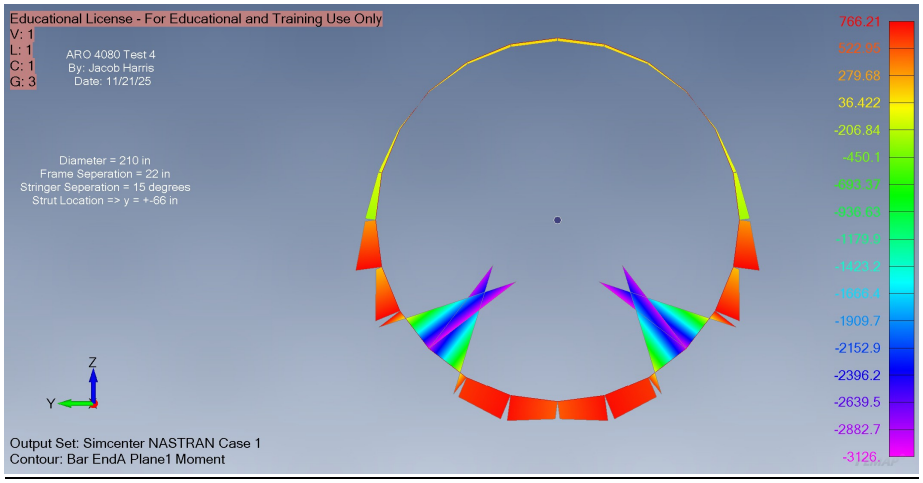


Figure 3.4.3: FEMAP FEM Bending Moments in Frames

3.5 Axial Load, Shear, Moment Diagrams in Floor Beams

The max Axial, Shear, and Moments are -201 lb, 390 lb, and -7245 in-lb respectively, located at the point where the struts meet the frames for the Shear and the Center for Axial and Moments.

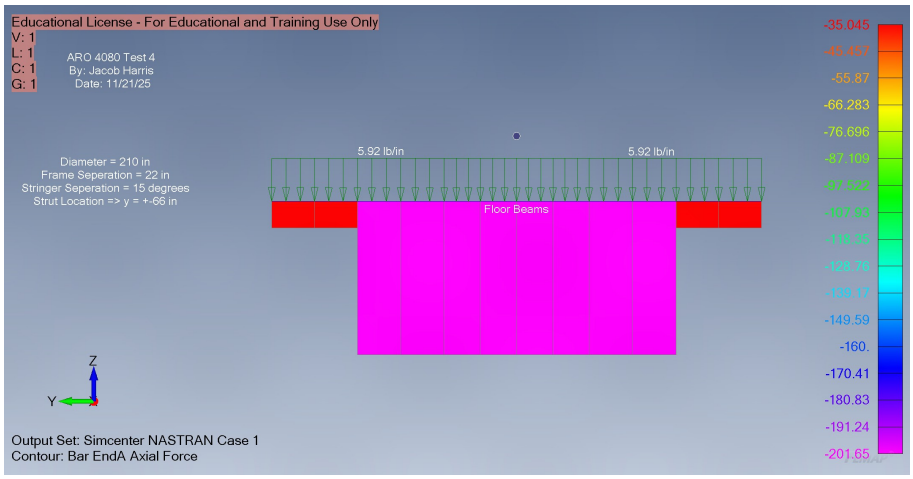


Figure 3.5.1: FEMAP FEM Axial Force in Floor Beams

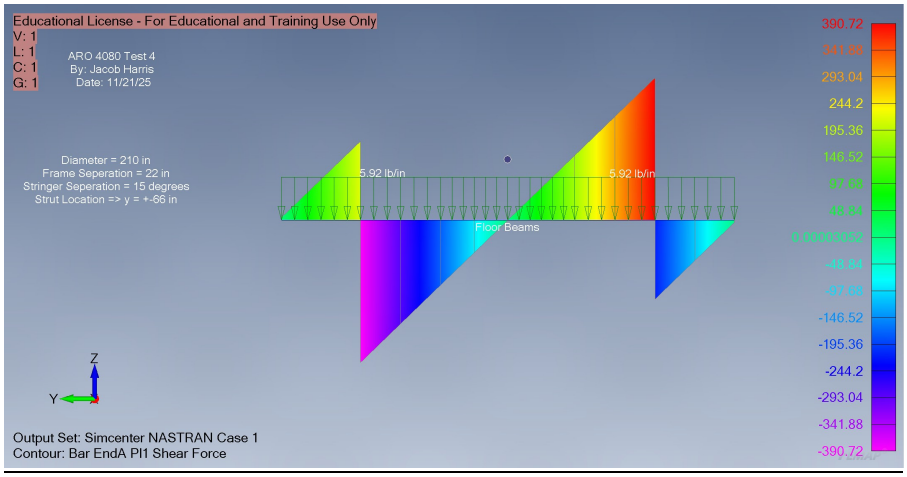


Figure 3.5.2: FEMAP FEM Shear in Floor Beams

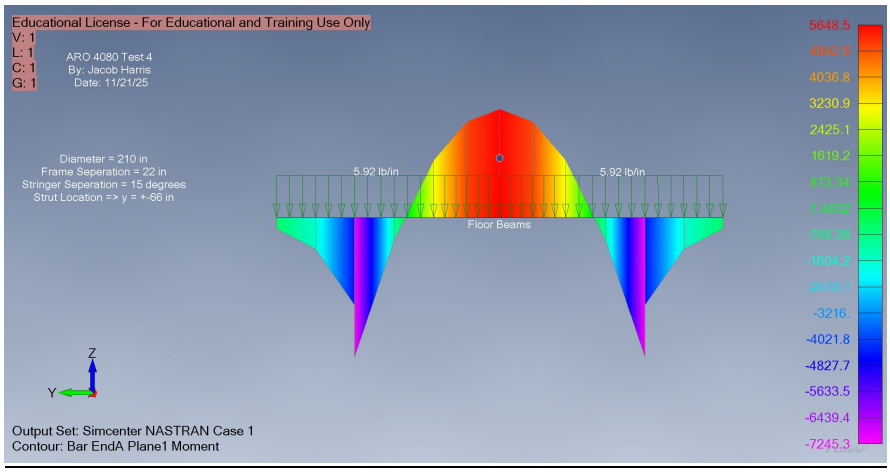


Figure 3.5.3: FEMAP FEM Bending Moment in Floor Beams

3.6 Axial Load, Shear, Moment Diagrams in Struts

The max Axial, Shear, and Moments are -626 lb, 59.4 lb, and 2706 in-lb respectively, located at the point where the struts meet the floor.

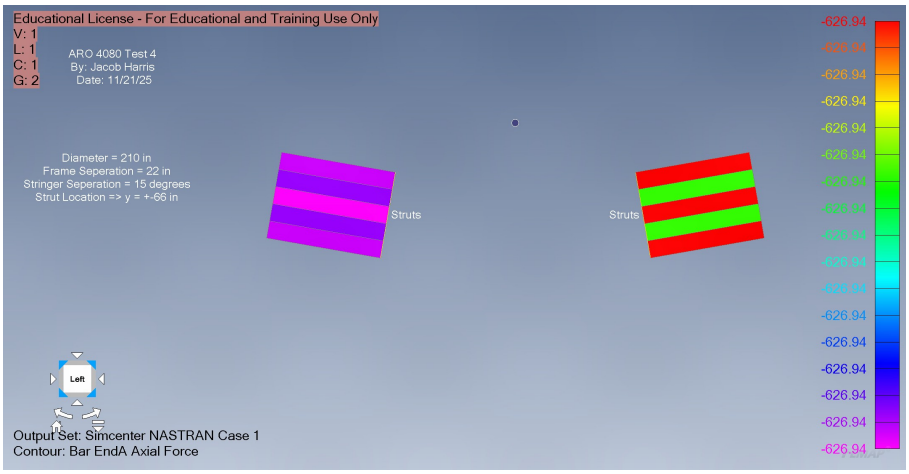


Figure 3.6.1: FEMAP FEM Axial Force in Struts

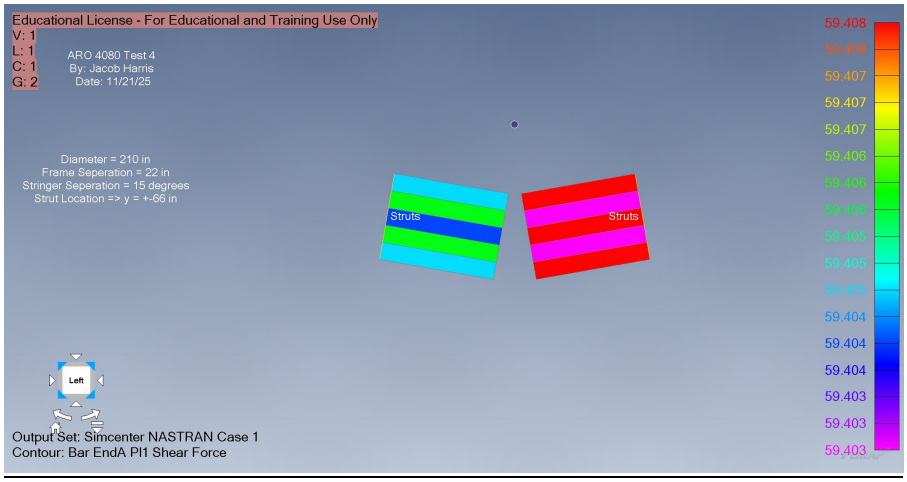


Figure 3.6.2: FEMAP FEM Shear in Struts

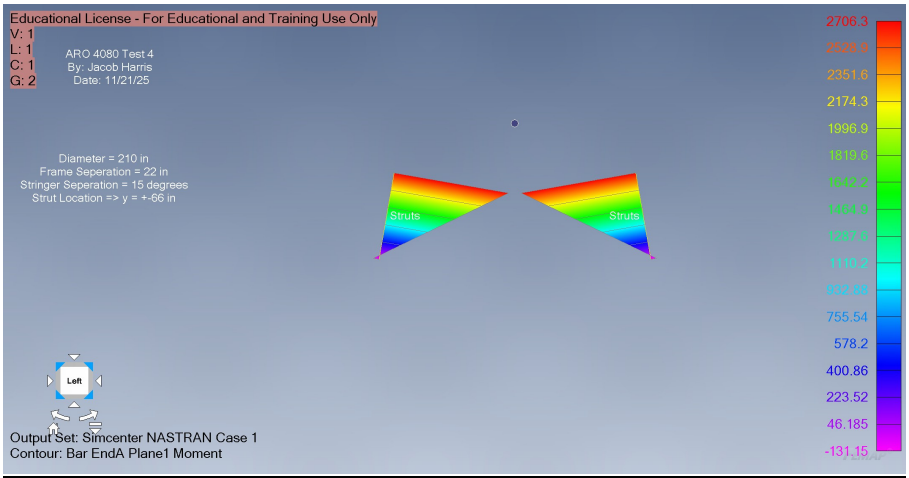


Figure 3.6.3: FEMAP FEM Bending Moments in Struts

4.0 Conclusion

4.1 What I did/What it Means

In this project I modeled one fuselage frame with an internal floor and tried to see how floor beam load actually travels through the structure. In FEMAP I built a circular frame with floor beams across the bottom, two diagonal struts that connect the floor back into the frame, and stringers that run around the frame. I also added skin panels so the cylinder isn't just a ring by itself. On each side of the main frame I put dummy frames and supported them with fixed breathing wagon-wheel constraints, so the main frame is held in place but can still expand radially. Then I applied a 1200 lb uniformly distributed load along the floor beams (5.92 lb/in) and ran a static analysis. After the run, I looked at the deformed shape, displacements of the floor and frame, SPC forces at the wagon wheels, and the internal forces and moments in the floor beams, frame, and struts.

What this really shows is the load path. The distributed load bends the floor beams down, the beams push into the struts, and the struts push into the frame in compression. The ring frame and skin spread that load around the circumference. When I sum the SPC forces they line up with the applied 1200 lb, so that gave me some confidence the model is behaving reasonably.

4.2 What I Learned

I learned how easy it is to mess things up with one wrong constraint or a missing connection. A small modeling mistake could either make the structure way too flexible or completely break it. I learned a lot of features that FEMAP has that were useful, even though some of them are deeply buried in tabs and not very intuitive or user friendly. But most of all, I learned how to properly analyze a fuselage frame.

4.3 Difficulty

I don't think this was a very difficult project, but it was extremely time-consuming. Mostly due lack of user friendliness that FEMAP has with its tools. But overall, the assignment itself was rather simple to understand, but it was just the amount of time it took to take that task is what made it challenging.

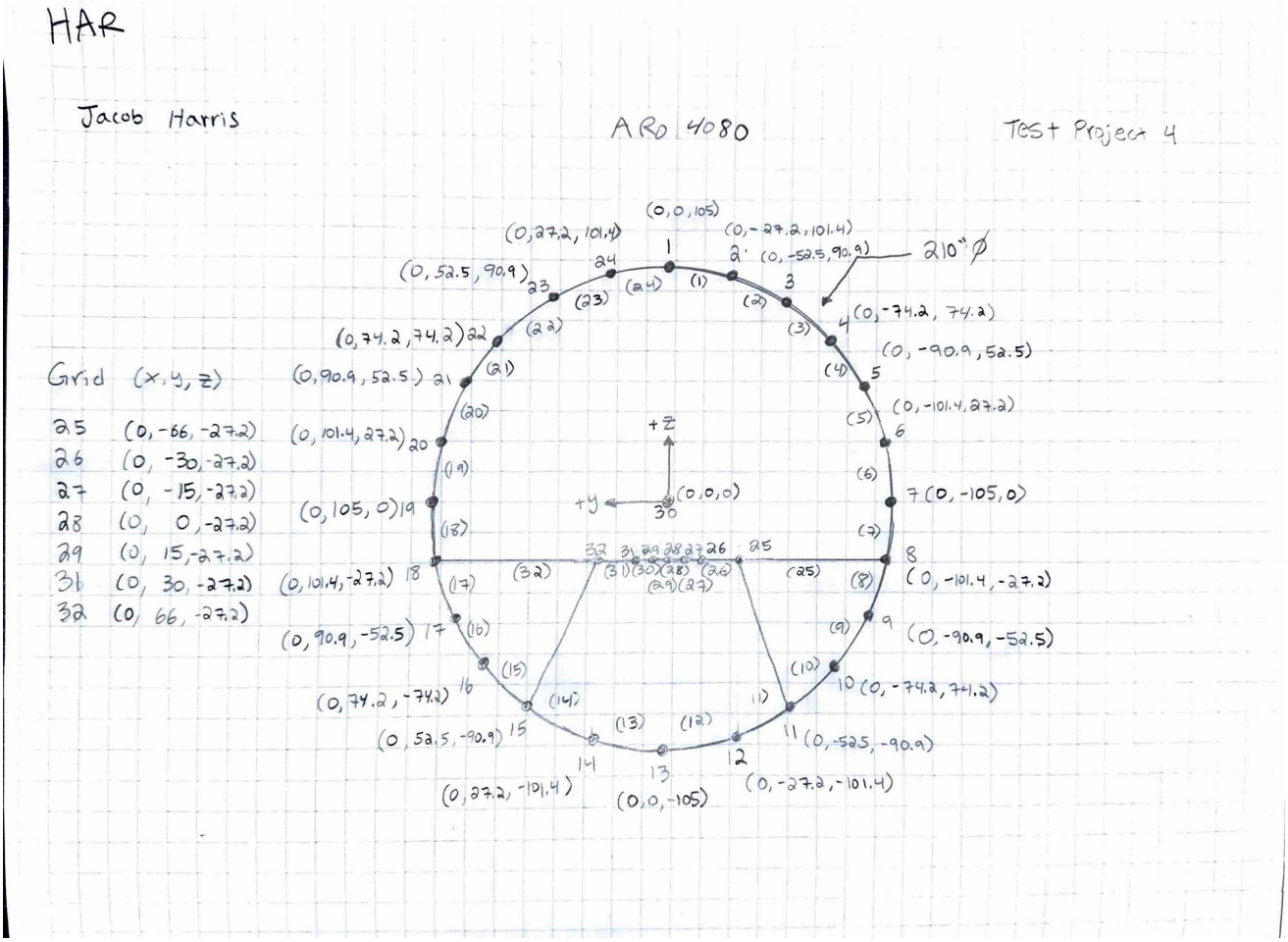
4.4 Usefulness of This Project

I think this project is very useful for someone who wants to be in this field, it shows exactly how to analyze a fuselage frame which we can use as a basis to analyze a whole fuselage. Not to mention it was a very fun alternative to a simple in class test.

4.5 Effectiveness of FEMAP

Now that I have gotten more comfortable with FEMAP, I understand now that it is extremely valuable regarding stress analysis. But you must really be careful in creating FEM's because it is super easy to mess something up and not realize it, and then you'll be spending the next day and a half trying to figure out what bonehead mistake you made, just to realize you had duplicate elements.

A FEM Preliminary Sketch



B Bulk Data File of FEM

```
INIT MASTER(S)
NASTRAN SYSTEM(442)=-1,SYSTEM(319)=1
ID Test4,FEMAP
SOL SESTATIC
CEND
TITLE = Simcenter Nastran Static Analysis Set
ECHO = NONE
DISPLACEMENT(PRINT) = ALL
SPCFORCE(PRINT) = ALL
OLOAD(PRINT) = ALL
MPCFORCE(PRINT) = ALL
GPFORCE(PRINT) = ALL
FORCE(PRINT,CORNER) = ALL
STRESS(PRINT,CORNER) = ALL
SPC = 1
LOAD = 1
BEGIN BULK
$ ************************************************************************
$   Written by : Femap
$   Version    : 2412.0
$   Translator : Simcenter Nastran
$   From Model : C:\Users\jacob\OneDrive\Desktop\FEMAP\ARO4080\Test4\Test4.modfem
$   Date       : Fri Nov 21 19:26:23 2025
$   Output To  : C:\Users\jacob\OneDrive\Desktop\FEMAP\ARO4080\Test4\
$ ************************************************************************
$
PARAM,PRGPST,NO
PARAM,POST,-1
PARAM,OGEOM,NO
PARAM,AUTOSPC,YES
PARAM,K6ROT,100.
PARAM,GRDPNT,0
CORD2C         1       0      0.      0.      0.      0.      0.      1.+FEMAPC1
+FEMAPC1      1.      0.      1.
CORD2S         2       0      0.      0.      0.      0.      0.      1.+FEMAPC2
+FEMAPC2      1.      0.      1.
$ Femap Load Set 1 : Load
PLOAD1         1      51     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      52     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      53     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      54     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      55     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      56     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      57     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      58     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      59     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      60     FYE      FR      0.   -5.92      1.   -5.92
PLOAD1         1      61     FYE      FR      0.   -5.92      1.   -5.92
```

```
PLOAD1          1      62     FYE      FR      0.   -5.92      1.   -5.92
$ Femap Constraint Set 1 : Breathing Wagon Wheel
SPC1            1  123456      74
SPC1            1  123456     118
$ Femap Property 1 : Frame
PBAR            1       1      .5      5.      .5      0.      0.        +
+              0.      0.      0.      0.      0.      0.      0.      0.+
+                              .05
$ Femap Property 2 : Struts
PBAR            2       1      .4      4.      .4     .04      0.        +
+              0.      0.      0.      0.      0.      0.      0.      0.+
+                              0.
$ Femap Property 3 : Floor
PBAR            3       1      .6      6.    .006     .06      0.        +
+              0.      0.      0.      0.      0.      0.      0.      0.+
+                              0.
$ Femap Property 4 : Stringers
PROD            4       1      .3     .03      0.      0.
$ Femap Property 5 : Skin
PSHELL          5       1    .063       1               1              0.
$ Femap Property 6 : BreathingWagonWheel
PBAR            6       1    .002 200000.     20.      .2      0.        +
+              0.      0.      0.      0.      0.      0.      0.      0.+
+                              0.
$ Femap Material 1 : Aluminum
MAT1            1    1.+73800000.     .33      0.      0.      0.
GRID            1       0      0.      0.    105.       0
GRID            2       0      0. -27.176101.4222       0
GRID            3       0      0.   -52.590.93267       0
GRID            4       0      0.-74.246274.24621       0
GRID            5       0      0.-90.9327    52.5       0
GRID            6       0      0.-101.422  27.176       0
GRID            7       0      0.   -105.      0.       0
GRID            8       0      0.-101.422 -27.176       0
GRID            9       0      0.-90.9327   -52.5       0
GRID           10       0      0.-74.2462-74.2462       0
GRID           11       0      0.   -52.5-90.9327       0
GRID           12       0      0. -27.176-101.422       0
GRID           13       0      0.      0.   -105.       0
GRID           14       0      0.  27.176-101.422       0
GRID           15       0      0.    52.5-90.9327       0
GRID           16       0      0.74.24621-74.2462       0
GRID           17       0      0.90.93267   -52.5       0
GRID           18       0      0.101.4222 -27.176       0
GRID           19       0      0.    105.      0.       0
GRID           20       0      0.101.4222  27.176       0
GRID           21       0      0.90.93267    52.5       0
GRID           22       0      0.74.2462174.24621       0
GRID           23       0      0.    52.590.93267       0
GRID           24       0      0.  27.176101.4222       0
```

```
GRID           25        0      0.    -66.   -27.2       0
GRID           26        0      0.    -30.   -27.2       0
GRID           27        0      0.    -15.   -27.2       0
GRID           28        0      0.      0.   -27.2       0
GRID           29        0      0.     15.   -27.2       0
GRID           30        0      0.      0.      0.       0
GRID           31        0      0.     30.   -27.2       0
GRID           32        0      0.     66.   -27.2       0
GRID           33        0      0.67.64924-36.6092       0
GRID           34        0      0.69.29848-46.0185       0
GRID           35        0      0.70.94773-55.4277       0
GRID           36        0      0.72.59697 -64.837       0
GRID           37        0      0.-67.6492-36.6092       0
GRID           38        0      0.-69.2985-46.0185       0
GRID           39        0      0.-70.9477-55.4277       0
GRID           40        0      0. -72.597 -64.837       0
GRID           41        0      0.83.71111 -27.188       0
GRID           42        0      0.     48.   -27.2       0
GRID           43        0      0.    -48.   -27.2       0
GRID           44        0      0.-83.7111 -27.188       0
GRID           45        0     22.      0.    105.       0
GRID           46        0     22. -27.176101.4222       0
GRID           47        0     22.   -52.590.93267       0
GRID           48        0     22.-74.246274.24621       0
GRID           49        0     22.-90.9327    52.5       0
GRID           50        0     22.-101.422  27.176       0
GRID           51        0     22.   -105.      0.       0
GRID           52        0     22.-101.422 -27.176       0
GRID           53        0     22.-90.9327   -52.5       0
GRID           54        0     22.-74.2462-74.2462       0
GRID           55        0     22.   -52.5-90.9327       0
GRID           56        0     22. -27.176-101.422       0
GRID           57        0     22.      0.   -105.       0
GRID           58        0     22.  27.176-101.422       0
GRID           59        0     22.    52.5-90.9327       0
GRID           60        0     22.74.24621-74.2462       0
GRID           61        0     22.90.93267   -52.5       0
GRID           62        0     22.101.4222 -27.176       0
GRID           63        0     22.    105.      0.       0
GRID           64        0     22.101.4222  27.176       0
GRID           65        0     22.90.93267    52.5       0
GRID           66        0     22.74.2462174.24621       0
GRID           67        0     22.    52.590.93267       0
GRID           68        0     22.  27.176101.4222       0
GRID           69        0     22.    -66.   -27.2       0
GRID           70        0     22.    -30.   -27.2       0
GRID           71        0     22.    -15.   -27.2       0
GRID           72        0     22.      0.   -27.2       0
GRID           73        0     22.     15.   -27.2       0
GRID           74        0     22.      0.      0.       0
```

```
GRID          75       0     22.     30.   -27.2       0
GRID          76       0     22.     66.   -27.2       0
GRID          77       0     22.67.64924-36.6092       0
GRID          78       0     22.69.29848-46.0185       0
GRID          79       0     22.70.94773-55.4277       0
GRID          80       0     22.72.59697 -64.837       0
GRID          81       0     22.-67.6492-36.6092       0
GRID          82       0     22.-69.2985-46.0185       0
GRID          83       0     22.-70.9477-55.4277       0
GRID          84       0     22. -72.597 -64.837       0
GRID          85       0     22.83.71111 -27.188       0
GRID          86       0     22.     48.   -27.2       0
GRID          87       0     22.    -48.   -27.2       0
GRID          88       0     22.-83.7111 -27.188       0
GRID          89       0    -22.      0.    105.       0
GRID          90       0    -22. -27.176101.4222       0
GRID          91       0    -22.   -52.590.93267       0
GRID          92       0    -22.-74.246274.24621       0
GRID          93       0    -22.-90.9327    52.5       0
GRID          94       0    -22.-101.422  27.176       0
GRID          95       0    -22.   -105.      0.       0
GRID          96       0    -22.-101.422 -27.176       0
GRID          97       0    -22.-90.9327   -52.5       0
GRID          98       0    -22.-74.2462-74.2462       0
GRID          99       0    -22.   -52.5-90.9327       0
GRID         100       0    -22. -27.176-101.422       0
GRID         101       0    -22.      0.   -105.       0
GRID         102       0    -22.  27.176-101.422       0
GRID         103       0    -22.    52.5-90.9327       0
GRID         104       0    -22.74.24621-74.2462       0
GRID         105       0    -22.90.93267   -52.5       0
GRID         106       0    -22.101.4222 -27.176       0
GRID         107       0    -22.    105.      0.       0
GRID         108       0    -22.101.4222  27.176       0
GRID         109       0    -22.90.93267    52.5       0
GRID         110       0    -22.74.2462174.24621       0
GRID         111       0    -22.    52.590.93267       0
GRID         112       0    -22.  27.176101.4222       0
GRID         113       0    -22.    -66.   -27.2       0
GRID         114       0    -22.    -30.   -27.2       0
GRID         115       0    -22.    -15.   -27.2       0
GRID         116       0    -22.      0.   -27.2       0
GRID         117       0    -22.     15.   -27.2       0
GRID         118       0    -22.      0.      0.       0
GRID         119       0    -22.     30.   -27.2       0
GRID         120       0    -22.     66.   -27.2       0
GRID         121       0    -22.67.64924-36.6092       0
GRID         122       0    -22.69.29848-46.0185       0
GRID         123       0    -22.70.94773-55.4277       0
GRID         124       0    -22.72.59697 -64.837       0
```

```
GRID          125       0    -22.-67.6492-36.6092       0
GRID          126       0    -22.-69.2985-46.0185       0
GRID          127       0    -22.-70.9477-55.4277       0
GRID          128       0    -22. -72.597 -64.837       0
GRID          129       0    -22.83.71111 -27.188       0
GRID          130       0    -22.     48.   -27.2       0
GRID          131       0    -22.    -48.   -27.2       0
GRID          132       0    -22.-83.7111 -27.188       0
CBAR            1       1       1       2      0.      0.     -1.
CBAR            2       1       2       3      0.      0.     -1.
CBAR            3       1       3       4      0.      0.     -1.
CBAR            4       1       4       5      0.      0.     -1.
CBAR            5       1       5       6      0.      0.     -1.
CBAR            6       1       6       7      0.      0.     -1.
CBAR            7       1       7       8      0.      0.     -1.
CBAR            8       1       8       9      0.      0.     -1.
CBAR            9       1       9      10      0.      0.     -1.
CBAR           10       1      10      11      0.      0.     -1.
CBAR           11       1      11      12      0.      0.     -1.
CBAR           12       1      12      13      0.      0.     -1.
CBAR           13       1      13      14      0.      0.     -1.
CBAR           14       1      14      15      0.      0.     -1.
CBAR           15       1      15      16      0.      0.     -1.
CBAR           16       1      16      17      0.      0.     -1.
CBAR           17       1      17      18      0.      0.     -1.
CBAR           18       1      18      19      0.      0.     -1.
CBAR           19       1      19      20      0.      0.     -1.
CBAR           20       1      20      21      0.      0.     -1.
CBAR           21       1      21      22      0.      0.     -1.
CBAR           22       1      22      23      0.      0.     -1.
CBAR           23       1      23      24      0.      0.     -1.
CBAR           24       1      24       1      0.      0.     -1.
CBAR           33       2      32      33      30
CBAR           34       2      25      37      30
CBAR           35       2      33      34      30
CBAR           36       2      34      35      30
CBAR           37       2      35      36      30
CBAR           38       2      36      16      30
CBAR           39       2      37      38      30
CBAR           40       2      38      39      30
CBAR           41       2      39      40      30
CBAR           42       2      40      10      30
CBAR           51       3      18      41       1
CBAR           52       3      32      42       1
CBAR           53       3      31      29       1
CBAR           54       3      29      28       1
CBAR           55       3      28      27       1
CBAR           56       3      27      26       1
CBAR           57       3      26      43       1
CBAR           58       3      25      44       1
```

```
CBAR          59       3      41      32       1
CBAR          60       3      42      31       1
CBAR          61       3      43      25       1
CBAR          62       3      44       8       1
CBAR          63       1      45      46      0.      0.     -1.
CBAR          64       1      46      47      0.      0.     -1.
CBAR          65       1      47      48      0.      0.     -1.
CBAR          66       1      48      49      0.      0.     -1.
CBAR          67       1      49      50      0.      0.     -1.
CBAR          68       1      50      51      0.      0.     -1.
CBAR          69       1      51      52      0.      0.     -1.
CBAR          70       1      52      53      0.      0.     -1.
CBAR          71       1      53      54      0.      0.     -1.
CBAR          72       1      54      55      0.      0.     -1.
CBAR          73       1      55      56      0.      0.     -1.
CBAR          74       1      56      57      0.      0.     -1.
CBAR          75       1      57      58      0.      0.     -1.
CBAR          76       1      58      59      0.      0.     -1.
CBAR          77       1      59      60      0.      0.     -1.
CBAR          78       1      60      61      0.      0.     -1.
CBAR          79       1      61      62      0.      0.     -1.
CBAR          80       1      62      63      0.      0.     -1.
CBAR          81       1      63      64      0.      0.     -1.
CBAR          82       1      64      65      0.      0.     -1.
CBAR          83       1      65      66      0.      0.     -1.
CBAR          84       1      66      67      0.      0.     -1.
CBAR          85       1      67      68      0.      0.     -1.
CBAR          86       1      68      45      0.      0.     -1.
CBAR          87       2      76      77      74
CBAR          88       2      69      81      74
CBAR          89       2      77      78      74
CBAR          90       2      78      79      74
CBAR          91       2      79      80      74
CBAR          92       2      80      60      74
CBAR          93       2      81      82      74
CBAR          94       2      82      83      74
CBAR          95       2      83      84      74
CBAR          96       2      84      54      74
CBAR          97       3      62      85      45
CBAR          98       3      76      86      45
CBAR          99       3      75      73      45
CBAR         100       3      73      72      45
CBAR         101       3      72      71      45
CBAR         102       3      71      70      45
CBAR         103       3      70      87      45
CBAR         104       3      69      88      45
CBAR         105       3      85      76      45
CBAR         106       3      86      75      45
CBAR         107       3      87      69      45
CBAR         108       3      88      52      45
```

```
CBAR         109       1      89      90      0.      0.     -1.
CBAR         110       1      90      91      0.      0.     -1.
CBAR         111       1      91      92      0.      0.     -1.
CBAR         112       1      92      93      0.      0.     -1.
CBAR         113       1      93      94      0.      0.     -1.
CBAR         114       1      94      95      0.      0.     -1.
CBAR         115       1      95      96      0.      0.     -1.
CBAR         116       1      96      97      0.      0.     -1.
CBAR         117       1      97      98      0.      0.     -1.
CBAR         118       1      98      99      0.      0.     -1.
CBAR         119       1      99     100      0.      0.     -1.
CBAR         120       1     100     101      0.      0.     -1.
CBAR         121       1     101     102      0.      0.     -1.
CBAR         122       1     102     103      0.      0.     -1.
CBAR         123       1     103     104      0.      0.     -1.
CBAR         124       1     104     105      0.      0.     -1.
CBAR         125       1     105     106      0.      0.     -1.
CBAR         126       1     106     107      0.      0.     -1.
CBAR         127       1     107     108      0.      0.     -1.
CBAR         128       1     108     109      0.      0.     -1.
CBAR         129       1     109     110      0.      0.     -1.
CBAR         130       1     110     111      0.      0.     -1.
CBAR         131       1     111     112      0.      0.     -1.
CBAR         132       1     112      89      0.      0.     -1.
CBAR         133       2     120     121     118
CBAR         134       2     113     125     118
CBAR         135       2     121     122     118
CBAR         136       2     122     123     118
CBAR         137       2     123     124     118
CBAR         138       2     124     104     118
CBAR         139       2     125     126     118
CBAR         140       2     126     127     118
CBAR         141       2     127     128     118
CBAR         142       2     128      98     118
CBAR         143       3     106     129      89
CBAR         144       3     120     130      89
CBAR         145       3     119     117      89
CBAR         146       3     117     116      89
CBAR         147       3     116     115      89
CBAR         148       3     115     114      89
CBAR         149       3     114     131      89
CBAR         150       3     113     132      89
CBAR         151       3     129     120      89
CBAR         152       3     130     119      89
CBAR         153       3     131     113      89
CBAR         154       3     132      96      89
CROD         155       4     111      23
CROD         156       4      23      67
CROD         157       4      68      24
CROD         158       4      24     112
```

```
CROD          159       4       45        1
CROD          160       4        1       89
CROD          161       4       46        2
CROD          162       4        2       90
CROD          163       4       47        3
CROD          164       4        3       91
CROD          165       4       48        4
CROD          166       4        4       92
CROD          167       4       49        5
CROD          168       4        5       93
CROD          169       4       50        6
CROD          170       4        6       94
CROD          171       4       51        7
CROD          172       4        7       95
CROD          173       4       52        8
CROD          174       4        8       96
CROD          175       4       53        9
CROD          176       4        9       97
CROD          177       4       54       10
CROD          178       4       10       98
CROD          179       4       55       11
CROD          180       4       11       99
CROD          181       4       56       12
CROD          182       4       12      100
CROD          183       4       57       13
CROD          184       4       13      101
CROD          185       4       58       14
CROD          186       4       14      102
CROD          187       4       59       15
CROD          188       4       15      103
CROD          189       4       60       16
CROD          190       4       16      104
CROD          191       4       61       17
CROD          192       4       17      105
CROD          193       4       62       18
CROD          194       4       18      106
CROD          195       4       63       19
CROD          196       4       19      107
CROD          197       4       64       20
CROD          198       4       20      108
CROD          199       4       65       21
CROD          200       4       21      109
CROD          201       4       66       22
CROD          202       4       22      110
CQUAD4        203       5       91        3       2       90
CQUAD4        204       5       92        4       3       91
CQUAD4        205       5       93        5       4       92
CQUAD4        206       5       94        6       5       93
CQUAD4        207       5       95        7       6       94
CQUAD4        208       5       96        8       7       95
```

```
CQUAD4      209       5      97       9       8      96
CQUAD4      210       5      98      10       9      97
CQUAD4      211       5      99      11      10      98
CQUAD4      212       5     100      12      11      99
CQUAD4      213       5     101      13      12     100
CQUAD4      214       5     102      14      13     101
CQUAD4      215       5     103      15      14     102
CQUAD4      216       5     104      16      15     103
CQUAD4      217       5     105      17      16     104
CQUAD4      218       5     106      18      17     105
CQUAD4      219       5     107      19      18     106
CQUAD4      220       5     108      20      19     107
CQUAD4      221       5     109      21      20     108
CQUAD4      222       5     110      22      21     109
CQUAD4      223       5     111      23      22     110
CQUAD4      224       5     112      24      23     111
CQUAD4      225       5      89       1      24     112
CQUAD4      226       5      90       2       1      89
CQUAD4      227       5       7      51      50       6
CQUAD4      228       5       8      52      51       7
CQUAD4      229       5       9      53      52       8
CQUAD4      230       5      10      54      53       9
CQUAD4      231       5      11      55      54      10
CQUAD4      232       5      12      56      55      11
CQUAD4      233       5      13      57      56      12
CQUAD4      234       5      14      58      57      13
CQUAD4      235       5      15      59      58      14
CQUAD4      236       5      16      60      59      15
CQUAD4      237       5      17      61      60      16
CQUAD4      238       5      18      62      61      17
CQUAD4      239       5      19      63      62      18
CQUAD4      240       5      20      64      63      19
CQUAD4      241       5      21      65      64      20
CQUAD4      242       5      22      66      65      21
CQUAD4      243       5      23      67      66      22
CQUAD4      244       5      24      68      67      23
CQUAD4      245       5       1      45      68      24
CQUAD4      246       5       2      46      45       1
CQUAD4      247       5       3      47      46       2
CQUAD4      248       5       4      48      47       3
CQUAD4      249       5       5      49      48       4
CQUAD4      250       5       6      50      49       5
CBAR        251       6      89     118      30
CBAR        252       6      90     118      30
CBAR        253       6      91     118      30
CBAR        254       6      92     118      30
CBAR        255       6      93     118      30
CBAR        256       6      94     118      30
CBAR        257       6      95     118      30
CBAR        258       6      96     118      30
```

```
CBAR          259       6      97     118      30
CBAR          260       6      98     118      30
CBAR          261       6      99     118      30
CBAR          262       6     100     118      30
CBAR          263       6     101     118      30
CBAR          264       6     102     118      30
CBAR          265       6     103     118      30
CBAR          266       6     104     118      30
CBAR          267       6     105     118      30
CBAR          268       6     106     118      30
CBAR          269       6     107     118      30
CBAR          270       6     108     118      30
CBAR          271       6     109     118      30
CBAR          272       6     110     118      30
CBAR          273       6     111     118      30
CBAR          274       6     112     118      30
CBAR          275       6      68      74      30
CBAR          276       6      67      74      30
CBAR          277       6      66      74      30
CBAR          278       6      65      74      30
CBAR          279       6      64      74      30
CBAR          280       6      63      74      30
CBAR          281       6      62      74      30
CBAR          282       6      61      74      30
CBAR          283       6      60      74      30
CBAR          284       6      59      74      30
CBAR          285       6      58      74      30
CBAR          286       6      57      74      30
CBAR          287       6      56      74      30
CBAR          288       6      55      74      30
CBAR          289       6      54      74      30
CBAR          290       6      53      74      30
CBAR          291       6      52      74      30
CBAR          292       6      51      74      30
CBAR          293       6      50      74      30
CBAR          294       6      49      74      30
CBAR          295       6      48      74      30
CBAR          296       6      47      74      30
CBAR          297       6      46      74      30
CBAR          298       6      45      74      30
ENDDATA 276101d6
```

References

[DLe5] Daryl Logan. A First Course in the Finite Element Method. 5th Edition. Cengage Learning. 2012.